ZONE	REV	DESCRIPTION	DATE	APPROVED DE	APPROVED PE	APPROVED CAD
	A	ORIGINAL RELEASE	06-12-23	LEILEI	STEVEN	JUSTIN

NOTES (UNLESS OTHERWISE SPECIFIED):

1. THIS DRAWING SPECIFIES THE REQUIREMENTS FOR A PRINTED WIRING BOARD IN ACCORDANCE. WITH SPECIFICATION IPC-6012 CLASS 2 (LATEST REVISION).

2. THE PWB MUST BE LEAD FREE ASSEMBLY PROCESS COMPATIBLE AND MUST BE ABLE TO HANDLE A MINIMUM OF 5 CYCLES AT 260 DEGREES CELSIUS FOR 10 SECONDS.

3. BASE MATERIAL - LAMINATE AND PREPREG SHALL MEET IPC-4101D-26, 83 or 98
   Tg - MUST BE GREATER THAN OR EQUAL TO 150 DEGREES CELSIUS.
   Td - MUST BE GREATER THAN OR EQUAL TO 330 DEGREES CELSIUS.

4. COPPER FOIL WEIGHT - SEE STACKUP DETAIL 'A'

5. CHARACTERISTIC IMPEDANCE - SEE DETAIL 'B'

6. MINIMUM CONDUCTIVE WIDTH/SPACING TO BE .006"/.004"

7. PLATING FINISH: A. BOTH SIDES ENIG; TO MEET THE REQUIREMENTS OF IPC-4552 (LATEST REVISION).

8. FAB VENDOR IS NOT ALLOWED TO USE ODB FOR FABRICATION. CAN BE USED ONLY FOR REFERENCE.

9. SOLDERMASK - TO MEET THE REQUIREMENTS OF IPC-SM-840E (OR LATEST REVISION).
   BLACK COLOR, BOTH SIDES. MODIFICATION OF SOLDERMASK IS NOT ALLOWED WITHOUT WRITTEN PERMISSION FROM NXP.
   TYPE: LPI OR EQUIVALENT.
   A. LOCATION = +/-.002" OF PLATED PADS.
   B. DIAMETER OR SIZE = +/-.002 OF ORIGINAL DATA

10. SILKSCREEN - WHITE EPOXY OR ACRYLIC INK, BOTH SIDES, NO SILKSCREEN ON ANY EXPOSED COPPER FEATURE.

11. ELECTRICAL TEST - 100% IPCD356. PCB FABRICATOR TO PERFORM A NET COMPARE AGAINST THE IPCD356 NETLIST PROVIDED BY NXP.

12. PRINTED WIRING BOARD IS TO BE INDIVIDUALLY BAGGED.

13. DFM CHECK MUST BE RUN ON BOARD DATA BEFORE BUILDING BOARDS, UNLESS PRIOR APPROVAL IS GIVEN IN WRITING BY NXP.

14. TEARDROPS MAY BE ADDED AT THE FAB HOUSE TO ALL SIGNAL LAYERS.

15. TWO SOLDER SAMPLES TO BE PROVIDED.

16. SUPPLIER MARKINGS - ON SECONDARY SIDE ONLY, WHERE SHOWN.
   - MUST BE UL RECOGNIZED AND MUST HAVE AN ID THAT CONFORMS TO UL94V-0

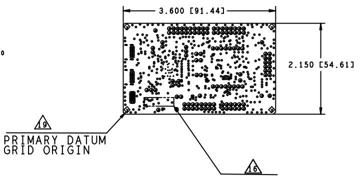
17. THE PWB WILL BE MARKED AS LEAD FREE BY USE OF AN INK STAMP (Pb)

18. THE PWB WILL BE MARKED AS LEAD FREE PROCESS COMPATIBLE BY USE OF AN INK STAMP (260°C)

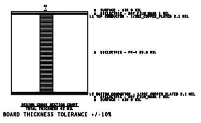
19. ALL PLATED AND NON-PLATED THROUGH HOLES ARE TO BE DRILLED AT PRIMARY DRILL STEP. ALL HOLE LOCATION TOLERANCES ARE TO BE +/-.002 IN REFERENCE TO THE PRIMARY DATUM UNLESS OTHERWISE SPECIFIED.

20. FINISHED PCB MUST BE PANELIZED FOR ASSEMBLY ACCORDING TO CONTRACT MANUFACTURERS REQUIREMENTS. THE ADDITION OF RAILS AND .125" NON-PLATED TOOLING HOLES ARE AT THE DISCRETION OF CONTRACT MANUFACTURER. PANELIZATION MUST BE APPROVED BY CONTRACT MANUFACTURER.

21. THE MANUFACTURE HAS THE OPTION TO ADD COPPER THIEVING ON OUTER AND INNER LAYERS, KEEP A MINIMUM DISTANCE OF .100" FROM ANY BOARD FEATURES.



DRILL CHART: TOP to BOTTOM					
ALL UNITS ARE IN MILS					
FIGURE	FINISHED_SIZE	TOLERANCE_DRILL	TOLERANCE_TRAVEL	PLATED	QTY
+	8.0	+0.0/-8.0	-	PLATED	19
○	10.0	+0.0/-10.0	-	PLATED	450
⬡	16.0	+2.0/-2.0	-	PLATED	32
②	28.0	+2.0/-2.0	-	PLATED	10
④	40.0	+3.0/-3.0	-	PLATED	123
◈	125.0	+2.0/-2.0	-	NON-PLATED	4
▭4	55.0x24.0	+2.0/-2.0	+2.0/-2.0	PLATED	4
▭	83.0x24.0	+2.0/-2.0	+2.0/-2.0	PLATED	4


SURFACE - AIR 0 MIL
DIELECTRIC - DRY FILM_MASK 1 MIL
L1 TOP CONDUCTOR - 1/2OZ_COPPER_PLATED 2.1 MIL
DIELECTRIC - FR-4 55.8 MIL
L2 BOTTOM CONDUCTOR - 1/2OZ COPPER PLATED 2.1 MIL
DIELECTRIC - DRY FILM_MASK 1 MIL
SURFACE - AIR 0 MIL
DESIGN CROSS SECTION CHART
TOTAL THICKNESS 62 MIL

BOARD THICKNESS TOLERANCE +/-10%

DETAIL A
LAYER STACKUP
SCALE: NONE

DETAIL B
IMPEDANCE REQUIREMENTS
IMPEDANCE TOLERANCE IS 10%

LAYERS	SINGLE ENDED		DIFFERENTIAL		
	TRACE WIDTH (MILS)	IMPEDANCE (OHMS)	TRACE WIDTH (MILS)	TRACE SPACING "AIR GAP" (MILS)	IMPEDANCE (OHMS)
L1_PS	6.00	50	8.67	4.33	90
L2_SS	6.00	50	8.67	4.33	90

ALL PARTS, MATERIALS AND FINISHED ASSEMBLY SHALL MEET THE RoHS COMMISSION DELEGATED DIRECTIVE (EU) 2015/863 OF 31 MARCH, 2015 AMENDING ANNEX II TO DIRECTIVE 2011/65/EU. A CERTIFICATE OF COMPLIANCE IS REQUIRED UPON REQUEST.

ALL PARTS, MATERIALS AND FINISHED ASSEMBLY SHALL NOT CONTAIN ANY OF THE SUBSTANCES OF VERY HIGH CONCERN (SVHC) ABOVE THE THRESHOLD VALUE PER THE CURRENT ECHA LIST OF SVHC'S AND WITH ANNEX XIV AND ANNEX XVII OF REACH DIRECTIVE 76/769/EEC. A CERTIFICATE OF COMPLIANCE IS REQUIRED UPON REQUEST.

___ COMPANY PUBLIC
X COMPANY INTERNAL
___ COMPANY CONFIDENTIAL

UNLESS OTHERWISE SPECIFIED
DIMENSIONS ARE IN INCHES
TOLERANCES ARE:
DECIMALS ANGLES
.XX .01 .0-30'
.XXX .005
√ RMS ALL MACHINED SURFACES
BREAK ALL SHARP EDGES AND CORNERS.
REMOVE BURRS.
UNDERLINED DIM. NOT TO SCALE.
THIRD ANGLE ORTHOGRAPHIC PROJECTION IS USED.

PART NO.
170-90829

THIS DOCUMENT CONTAINS INFORMATION PROPRIETARY TO NXP AND SHALL NOT BE USED FOR ENGINEERING DESIGN PROCUREMENT OR MANUFACTURE IN WHOLE OR IN PART WITHOUT THE CONSENT OF NXP.

APPROVALS	DATE
DRAWN JUSTIN XIANG	06-12-23
CHECKED STEVEN DING	06-12-23
DESIGN ENGINEER LEILEI DU	06-12-23

NXP SEMICONDUCTORS
6501 WILLIAM CANNON DRIVE WEST AUSTIN, TEXAS 78735 USA

TITLE:
PRINTED WIRING BOARD
FRDM-MCXA153

SIZE D
CAD FILE NAME LAY-90829
DWG. NO. FAB-90829
REV A
SCALE 1/1
DO NOT SCALE DRAWING
SHEET 1 OF 1